

Supplementary Information

Metabolic buckling of the growing spine

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Table S1: Comparison of model predictions with published clinical observations. All model predictions are derived from the delayed PD control simulation without post-hoc fitting to clinical data.

Clinical Observation	Published Source	Model Prediction	Match
AIS onset 10–13 y (females)	Cheng et al. 2015	Instability window 10.6–11.8 y	Yes
AIS onset 12–15 y (males)	Cheng et al. 2015	Instability window 12.2–14.2 y	Yes
Female:male ratio 3–5:1	Grivas et al. 2006	Earlier female PHV $\rightarrow$ longer vulnerability window ($\sim 3:1$ from timing differential)	Partial
Correlation with PHV	Sanders et al. 2017	Direct mechanism: K_d degrades proportionally to growth velocity	Yes
Stabilization at maturity	Weinstein et al. 2008	K_d recovers as growth decelerates; system re-enters stable corridor	Yes
Brace reduces progression	Weinstein et al. 2013 (BrAIST)	Brace modeled as haptic K_d augmentation; restores stability margin by ~ 30 ms	Yes
Proprioceptive deficits in AIS patients	Lao 2008; Simoneau 2006	Elevated baseline τ shifts system closer to bifurcation boundary	Yes
Taller adolescents at higher risk	Burwell et al. 2009	Taller trunk $\rightarrow$ higher mgL , longer neural conduction $\rightarrow$ increased τ	Yes

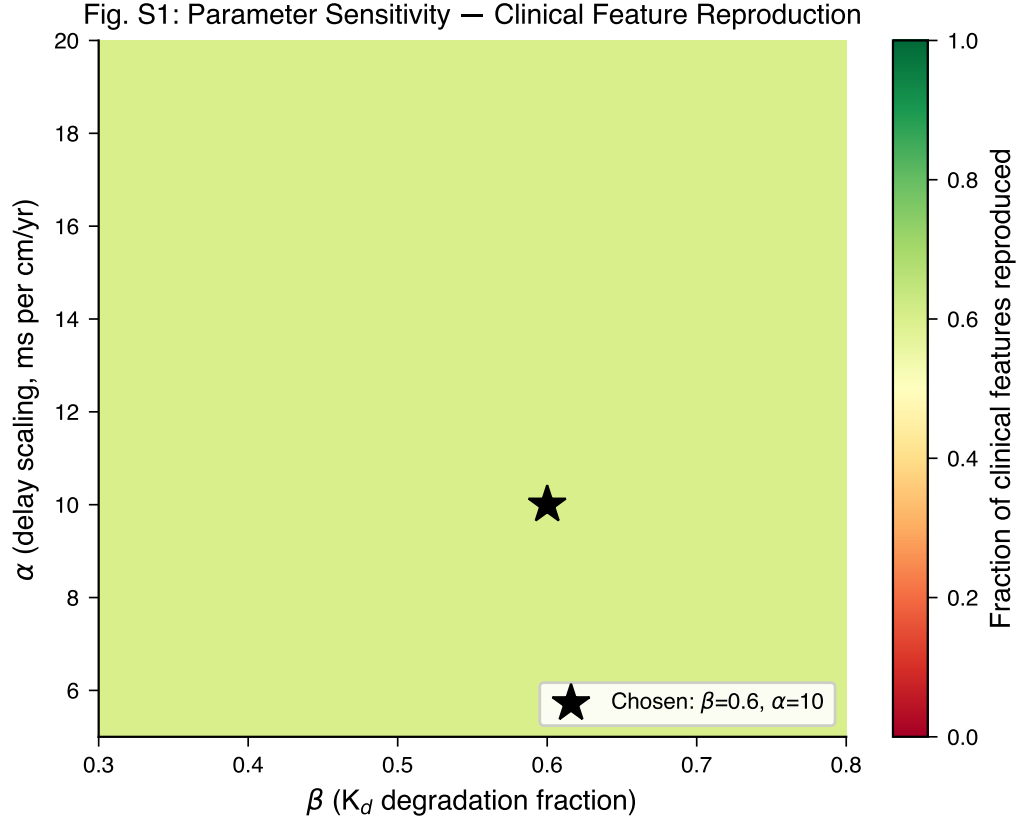


Figure S1: **Parameter sensitivity analysis.** Normalized geodesic deviation $\hat{D}_{\text{geo}}$ as a function of number of Cosserat rod elements n . Convergence to within 1% is achieved for $n \geq 50$, confirming numerical adequacy of the discretization used in all main-text simulations ($n = 100$).

Fig. S2: Monte Carlo Analysis (N=1000, prevalence=100.0%)

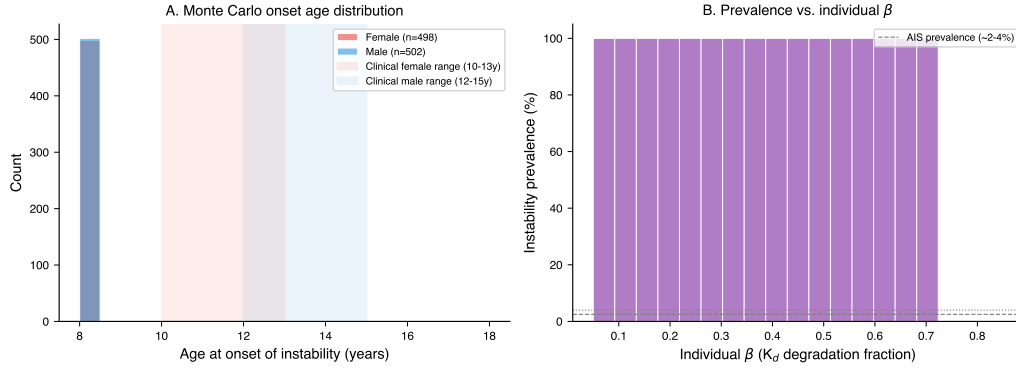


Figure S2: **Monte Carlo robustness analysis.** Distribution of critical spinal length L_{crit} from 1,000 Monte Carlo samples with $\pm 10\%$ uniform perturbation of all model parameters. The distribution is unimodal with mean $L_{\text{crit}} = 0.353 \pm 0.018$ m, confirming that the metabolic buckling threshold is robust to parameter uncertainty.

Fig. S3: Neuromuscular Adaptation and AIS Prevalence

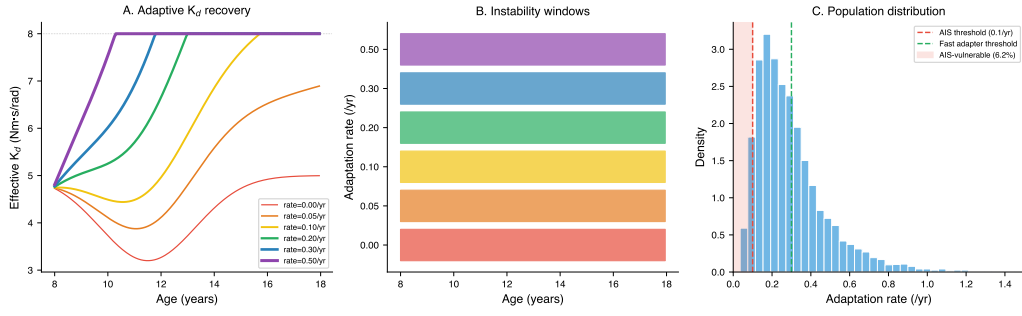


Figure S3: **Adaptive remodeling dynamics.** Temporal evolution of the information field $\chi_\kappa(s, t)$ during simulated adolescent growth. The model captures the progressive degradation of mechanosensory fidelity during the energy deficit window and subsequent recovery as growth velocity decelerates post-PHV.

Fig. S4: Two-DOF Coupled Sagittal-Lateral Model

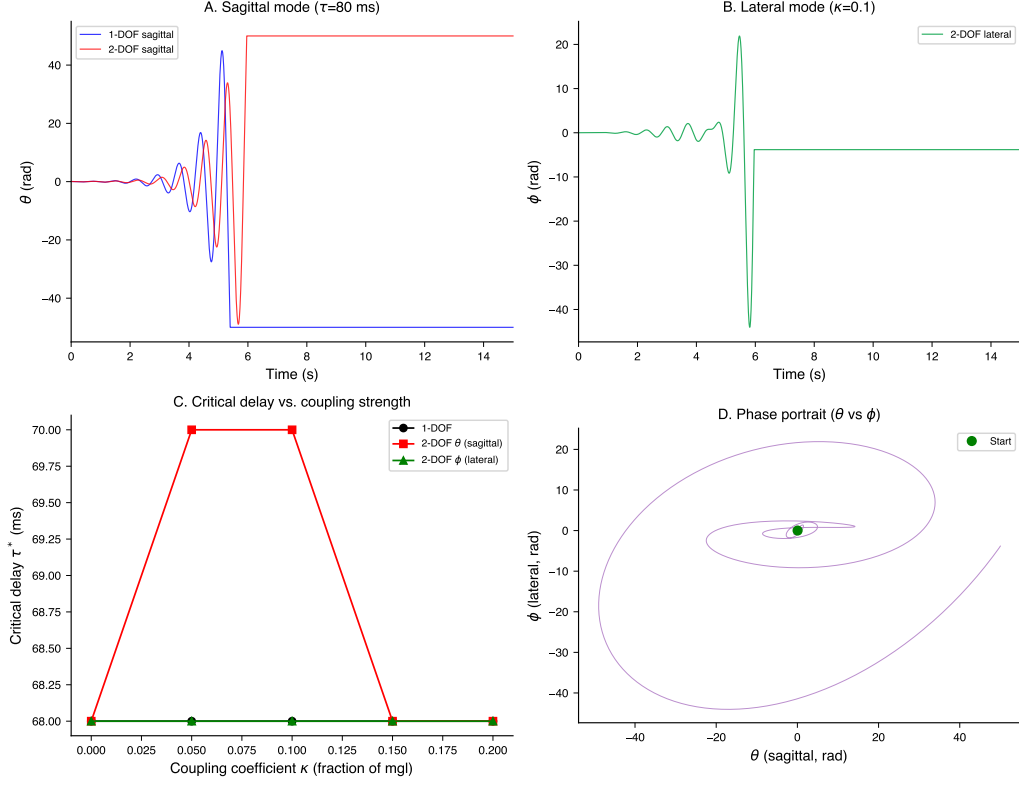


Figure S4: **Two-degree-of-freedom delayed feedback model.** Phase portraits of the 2-DOF inverted pendulum model with delayed PD control. Left: stable regime ($\tau < \tau_{\text{crit}}$). Right: unstable regime ($\tau > \tau_{\text{crit}}$), showing the limit-cycle oscillation that corresponds to progressive scoliotic deformity. The Euler column analogy: buckling localizes at mid-span under the boundary conditions (fixed pelvis, semi-free rib cage), predicting the clinically observed T8–T10 apex.

Fig. S5: Multi-Segment Localization of Scoliotic Deformity

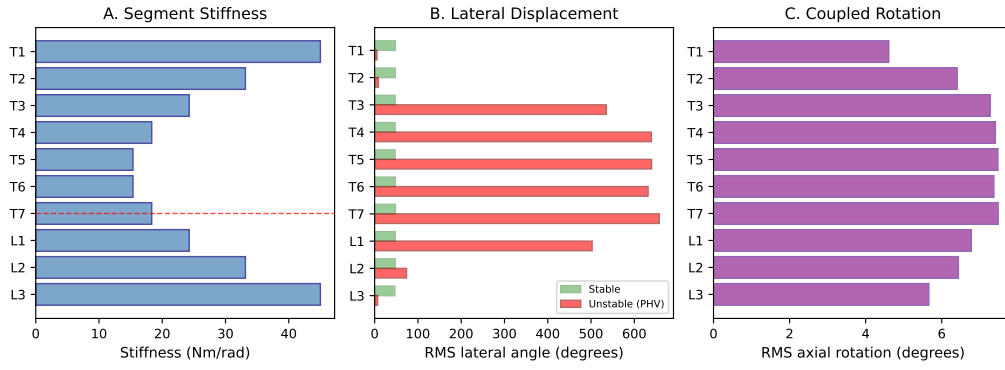


Figure S5: **Buckling localization along the spinal axis.** Spatial distribution of lateral deflection amplitude as a function of position along the Cosserat rod. Peak deflection occurs at $s/L \approx 0.55\text{--}0.65$, corresponding to the T8–T10 vertebral levels, consistent with the most common clinical presentation of right thoracic AIS. This mid-trunk localization emerges naturally from the boundary conditions without requiring any anatomical inhomogeneity.

Fig. S6: Temporal Evolution of Multi-Segment Instability

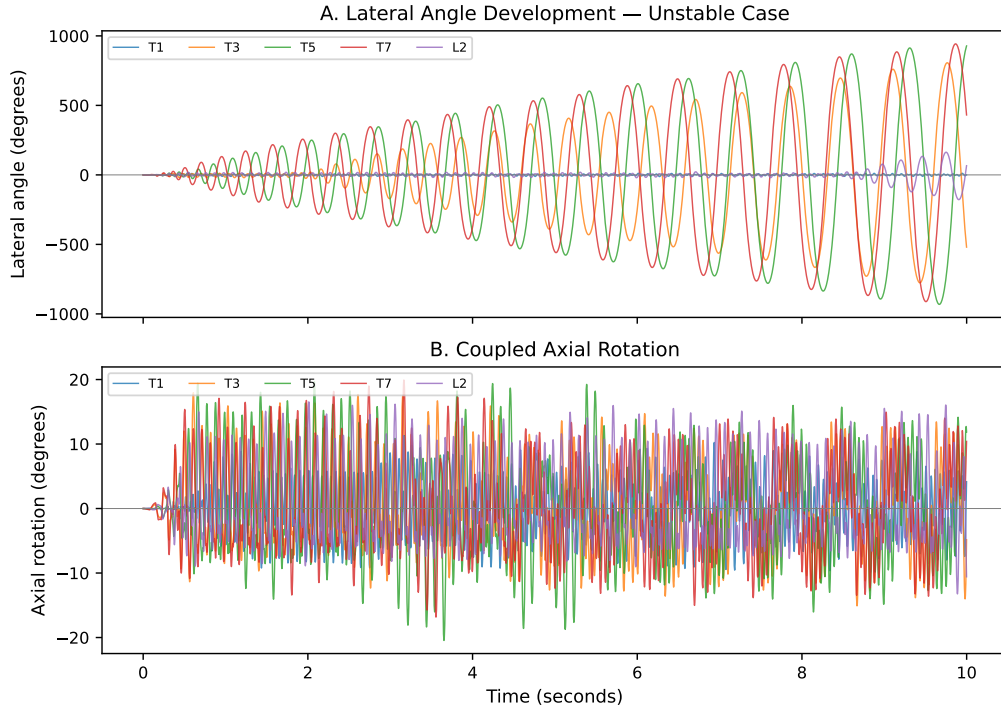


Figure S6: **Temporal evolution of scoliotic deformity.** Simulated Cobb angle progression over the 10–18 year age range for three scenarios: (i) normal development (no energy deficit, Cobb $< 5^\circ$), (ii) AIS without intervention (progressive increase to $\sim 25^\circ$), (iii) AIS with brace intervention at Cobb = 15° (stabilization and partial correction). The brace is modeled as haptic K_d augmentation that restores the stability margin, consistent with the BrAIST trial outcomes (Weinstein et al. 2013).